

DA1 - MINI PROJECT DOCUMENTATION TEMPLATE

Page 1: Cover Page

Title:

C Programming Mini Project – Digital Assignment

Project Title:

(Example: Movie Ticket Booking System)

Student Name:

Register Number:

Department:

Course Name:

Faculty Name:

Submission Date:

GitHub Repository Link:

Live Hosted Link (GitHub Pages / Replit):

1. Problem Statement (10–12 lines)

Explain:

- What problem your system solves
- Who will use this system
- Why it is useful
- What main features are implemented

Must be written in your own words.

2. Features Implemented

List clearly (bullet format):

- Add record
- Update record
- Delete record
- Search record
- File storage
- Report generation
- Input validation
- Any additional feature

Minimum 6 features required.

3. System Design

3.1 Structure Definition

Paste your struct here and explain each field:

```
struct Student {  
    int regNo;  
    char name[50];  
    float marks[3];  
};
```

Explain:

- Why you chose these fields
- What each field represents

3.2 Function List

Create table:

Function Name	Purpose	Inputs	Output
addRecord()	Adds new record	user input	none

searchRecord()	Finds record by ID	ID	record
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Minimum 5 user-defined functions required.

3.3 File Handling Design

Explain:

- File name used
- What data is stored in file
- File format (text/binary)
- Example of stored record

Example:

1001,Arun,80,75,90

4. Program Flow Explanation

Explain briefly:

- How menu works
- How adding works
- How searching works
- How updating/deleting works
- How reports are generated

You may include a flowchart (optional but encouraged).

5. Core Logic Explanation

Explain important logic:

Example:

- Seat booking validation logic
- Grade calculation logic
- Salary calculation logic
- Attendance percentage calculation

Write in simple explanation (no need to be very technical).

6. Testing and Validation

6.1 Test Case Table (Minimum 6)

TC No	Input	Expected Output	Actual Output	Result
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Must include:

- Normal test case
- Invalid input case
- Boundary case
- File not found case (if applicable)

6.2 Error Handling Implemented

Explain:

- What invalid inputs you handled
- How you handled them
- Example: negative amount, duplicate ID, invalid seat number, etc.

7. Screenshots

Insert screenshots:

1. Program running
2. Sample output
3. Live hosted page
4. GitHub repository page (optional)

Screenshots must be clear.

8. Deployment Explanation

Explain:

- How you compiled to WebAssembly (if used)
- Steps followed for GitHub Pages
- Any issues faced during deployment

9. Challenges Faced

Write at least 5–6 lines:

- One bug you faced
- How you solved it
- One concept that was difficult
- What you learned

10. AI Usage Declaration (

Answer clearly:

1. Did you use AI? (Yes/No)
2. For what purpose?
3. What modifications did you make?
4. What part did you debug yourself?

11. Complete Source Code (Compulsory Section)

Students must:

- Paste entire C program
- Proper indentation
- Include comments explaining logic
- No screenshots of code
- Must be copyable text

Example header format:

```
/*  
    Project Title: Movie Ticket Booking System  
    Student Name:  
    Register No:  
    */  
  
#include <stdio.h>  
#include <stdlib.h>
```

12. Conclusion

Write 5–6 lines:

- What you learned from this mini project
- Which C concepts were applied
- How this helped your understanding

⚠ Formatting Rules

- Font: Times New Roman / Calibri
- Size: 12
- Line spacing: 1.5
- Page numbers mandatory
- PDF only (No Word document)